

EXPERIMENT-04

Aim:

To analyse overfitting in a machine-learning model by comparing its training and test performance using the Breast Cancer dataset.

Algorithm:

```
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

overfit_model = DecisionTreeClassifier(random_state=42)
overfit_model.fit(X_train, y_train)

overfit_train_accuracy = accuracy_score(y_train, overfit_model.predict(X_train))
overfit_test_accuracy = accuracy_score(y_test, overfit_model.predict(X_test))

regularized_model = DecisionTreeClassifier(max_depth=3, random_state=42)
regularized_model.fit(X_train, y_train)

regularized_train_accuracy = accuracy_score(
```

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    y_train, regularized_model.predict(X_train)
)
regularized_test_accuracy = accuracy_score(
    y_test, regularized_model.predict(X_test)
)

print("OVERFITTING ANALYSIS\n")

print("1. Unrestricted Decision Tree")
print("Training Accuracy:", round(overfit_train_accuracy * 100, 2), "%")
print("Testing Accuracy :", round(overfit_test_accuracy * 100, 2), "%")

print("\n2. Controlled Decision Tree (max_depth = 3)")
print("Training Accuracy:", round(regularized_train_accuracy * 100, 2), "%")
print("Testing Accuracy :", round(regularized_test_accuracy * 100, 2), "%")

models = ["Unrestricted Tree", "Controlled Tree"]
training_scores = [overfit_train_accuracy * 100, regularized_train_accuracy * 100]
testing_scores = [overfit_test_accuracy * 100, regularized_test_accuracy * 100]

x = range(len(models))

plt.figure(figsize=(9, 5))
plt.bar([i - 0.2 for i in x], training_scores, width=0.4,
        label="Training Accuracy", color="skyblue")
plt.bar([i + 0.2 for i in x], testing_scores, width=0.4,

```

```
label="Testing Accuracy", color="orange")
```

```
plt.xticks(x, models)
```

```
plt.ylabel("Accuracy (%)")
```

```
plt.title("Overfitting Analysis")
```

```
plt.ylim(0, 105)
```

```
plt.legend()
```

```
plt.show()
```

CODE:

```
# Overfitting Analysis using Decision Tree
# Dataset: Breast Cancer dataset from scikit-learn

import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split dataset: 80% training, 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

# Model 1: Unrestricted tree - likely to overfit
overfit_model = DecisionTreeClassifier(random_state=42)
```

OUTPUT

